Forming Simultaneous Equations

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Written by: Mark Curtis
Reviewed by: Dan Finlay
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Forming Simultaneous Equations

How do I form simultaneous equations?

- Introduce **two letters**, x and y, to represent **two unknowns**
  - Make sure you know exactly what they stand for (and any units)
- Create **two different equations** from the words or contexts
  - 3 apples and 2 bananas cost \$1.80, while 5 apples and 1 banana cost \$2.30
    - $3x + 2y = 180$ and $5x + y = 230$
      x is the **price** of an apple, in **cents**
      y is the **price** of a banana, in **cents**
    - This question could also be done in dollars, \$
- **Solve** the equations **simultaneously**
- Give answers **in context** (relate them to the story, with units)
  - $x = 40$, $y = 30$
  - In context: an apple costs 40 cents and a banana costs 30 cents
- Some questions don't ask you to solve simultaneously, but you still need to
  - Two numbers have a **sum** of 19 and a **difference** of 5, what is their **product**?
    - $x + y = 19$ and $x - y = 5$
    - Solve simultaneously to get $x = 12$, $y = 7$
    - The product is $xy = 12 \times 7 = 84$

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Examiner Tips and Tricks

- Always check that you've answered the question! Sometimes finding x and y isn't the end

Worked Example

At a bakery a customer pays £9 in total for six bagels and twelve sausage rolls.

Another customer buys nine bagels and ten sausage rolls, which costs £12.30 in total.

Find the cost of 5 bagels and 15 sausage rolls.

Answer:

The two variables are the price of bagels, b, and the price of sausage rolls, s

Write an equation for the first customer's purchases, and label it equation 1

$$(1) \quad 6b + 12s = 9$$

Write an equation for the second customer's purchases, and label it equation 2

$$(2) \quad 9b + 10s = 12.3$$

We will choose to eliminate the b terms
Make the b terms equal by multiplying all parts of equation 1 by 3 and all parts of equation 2 by 2
Label these as equations 3 and 4

$$(1) \times 3 \quad 18b + 36s = 27 \qquad (3)$$
$$(2) \times 2 \quad 18b + 20s = 24.6 \qquad (4)$$

To eliminate b, subtract equation 4 from equation 3

$$(3) - (4) \quad 16s = 2.4$$

Solve for s

$$s = \frac{2.4}{16} = 0.15$$

Substitute this into either equation to find b, we will use equation 1

$$(1) \quad 6b + 12(0.15) = 9$$
$$6b + 1.8 = 9$$
$$6b = 7.2$$
$$b = 1.2$$

So sausage rolls cost £0.15 each and bagels cost £1.20 each

Use these values to find the price of 5 bagels and 15 sausage rolls

$$(5 \times 1.2) + (15 \times 0.15) = 8.25$$

£8.25

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